DEVOPS FINAL CAPSTONE · LOCAL DEMONSTRATION

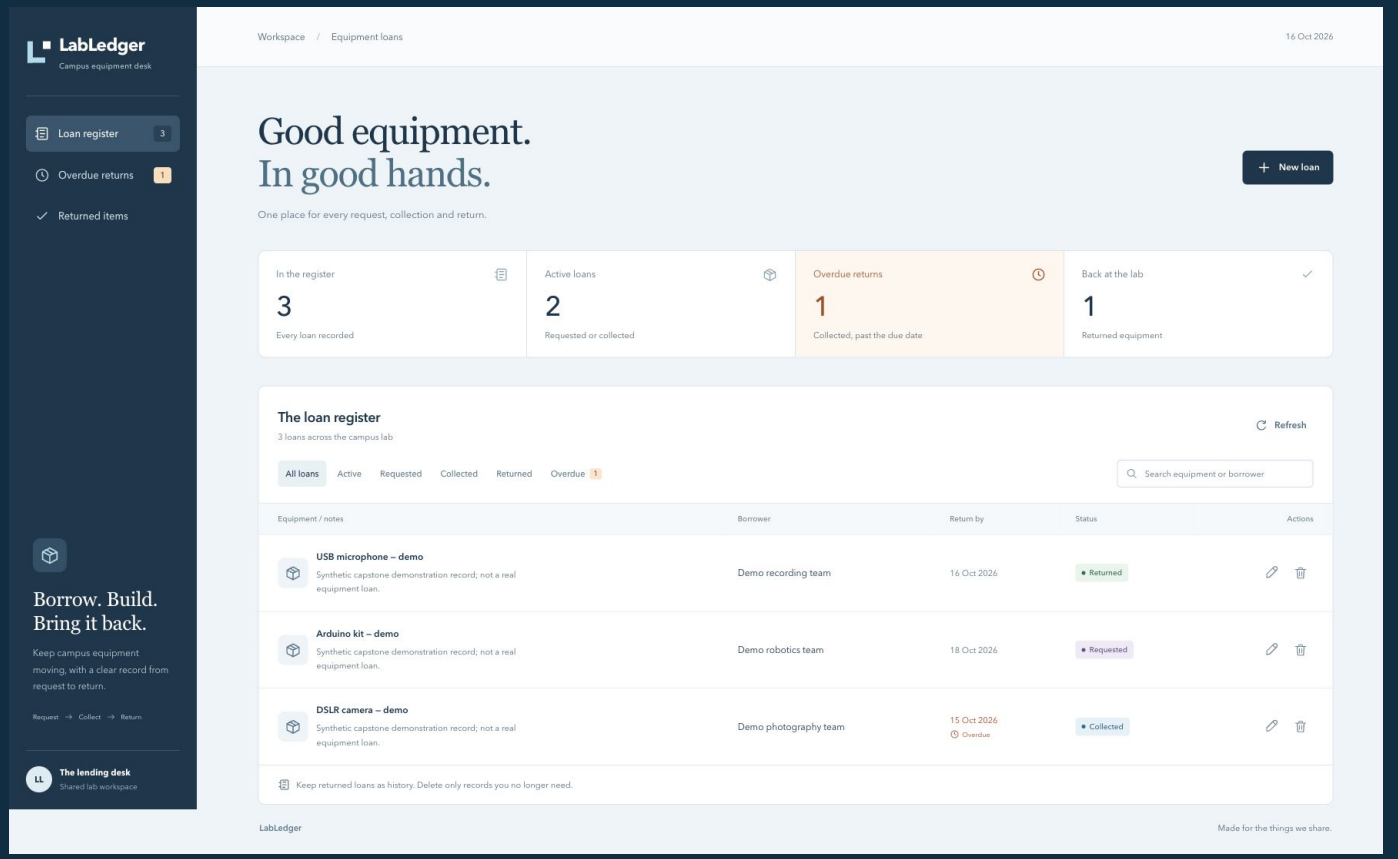
LabLedger

Shared equipment. Clear returns.

React · FastAPI · PostgreSQL
Docker · CI · Helm · Monitoring

VERIFIED LOCALLY

Cloud execution is still pending.



One clear equipment-loan lifecycle

Request → collect → return

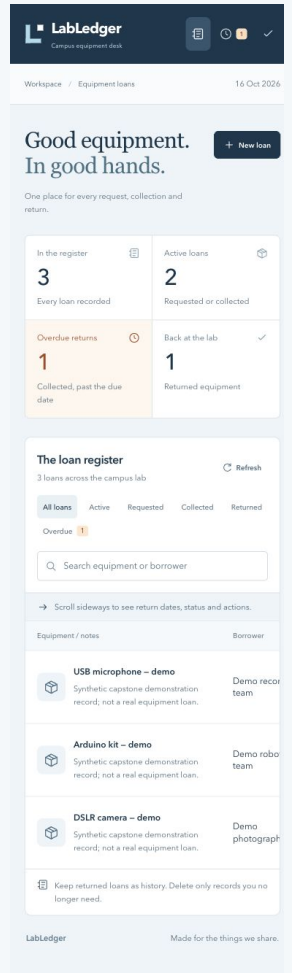
A lab desk records the borrower, equipment and return date.

Overdue has a precise meaning

Collected + due date before today in Asia/Kolkata. Requested loans are active, but not overdue.

Usable on small screens

Search, status filters, readable actions and an accessible dialog. Demonstration records are explicitly fictional.



Genuine School-profile mobile capture

Follow a request through the stack

React UI
Relative /api calls

→

Nginx / Ingress
/ and /api routes

→

FastAPI
Validation + metrics

→

PostgreSQL
Alembic + loans

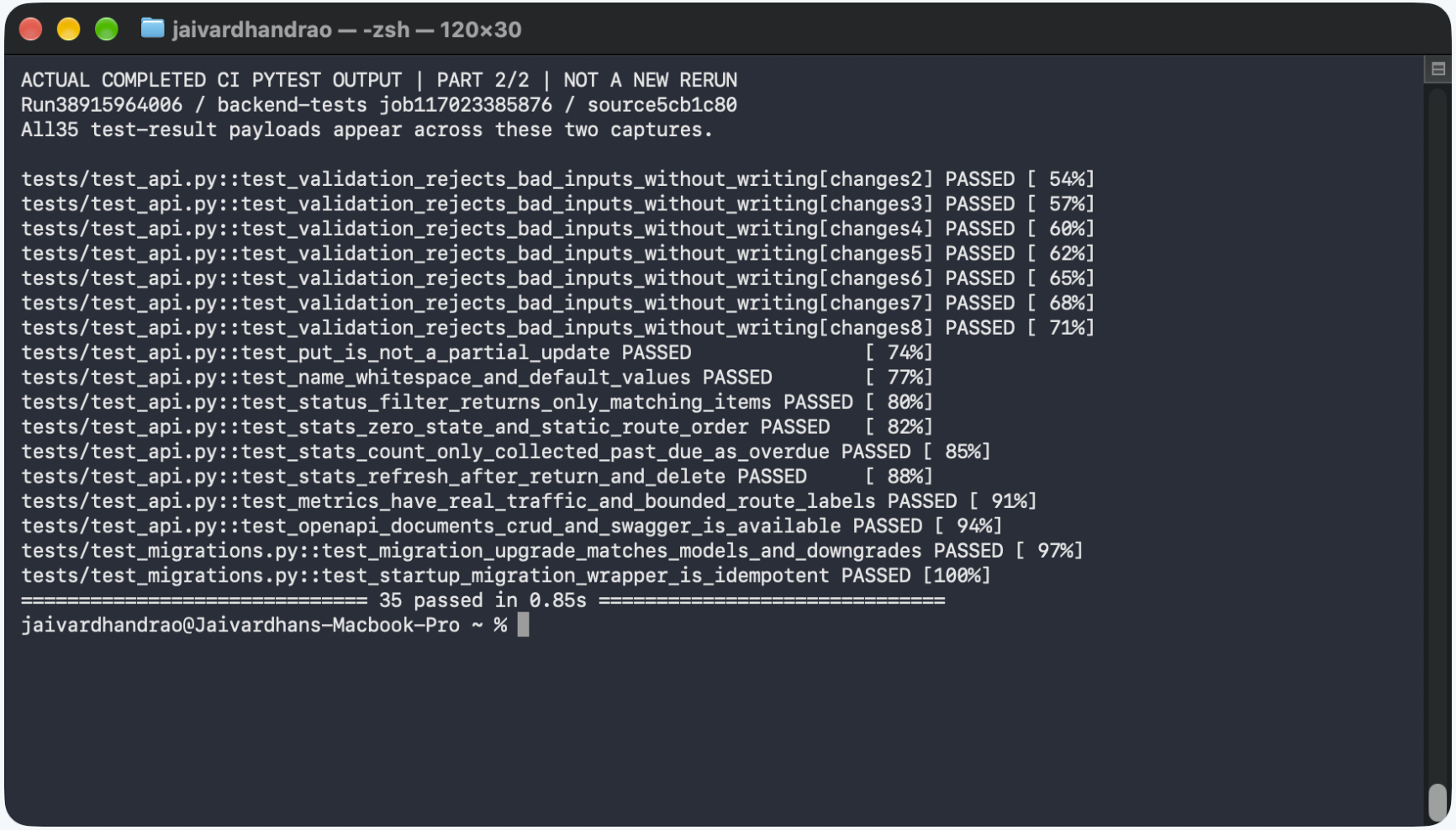
Compose localhost:3000 → frontend → backend → database

Kubernetes localhost:8088 → Ingress → ClusterIP Services → Pods

Prometheus → Grafana

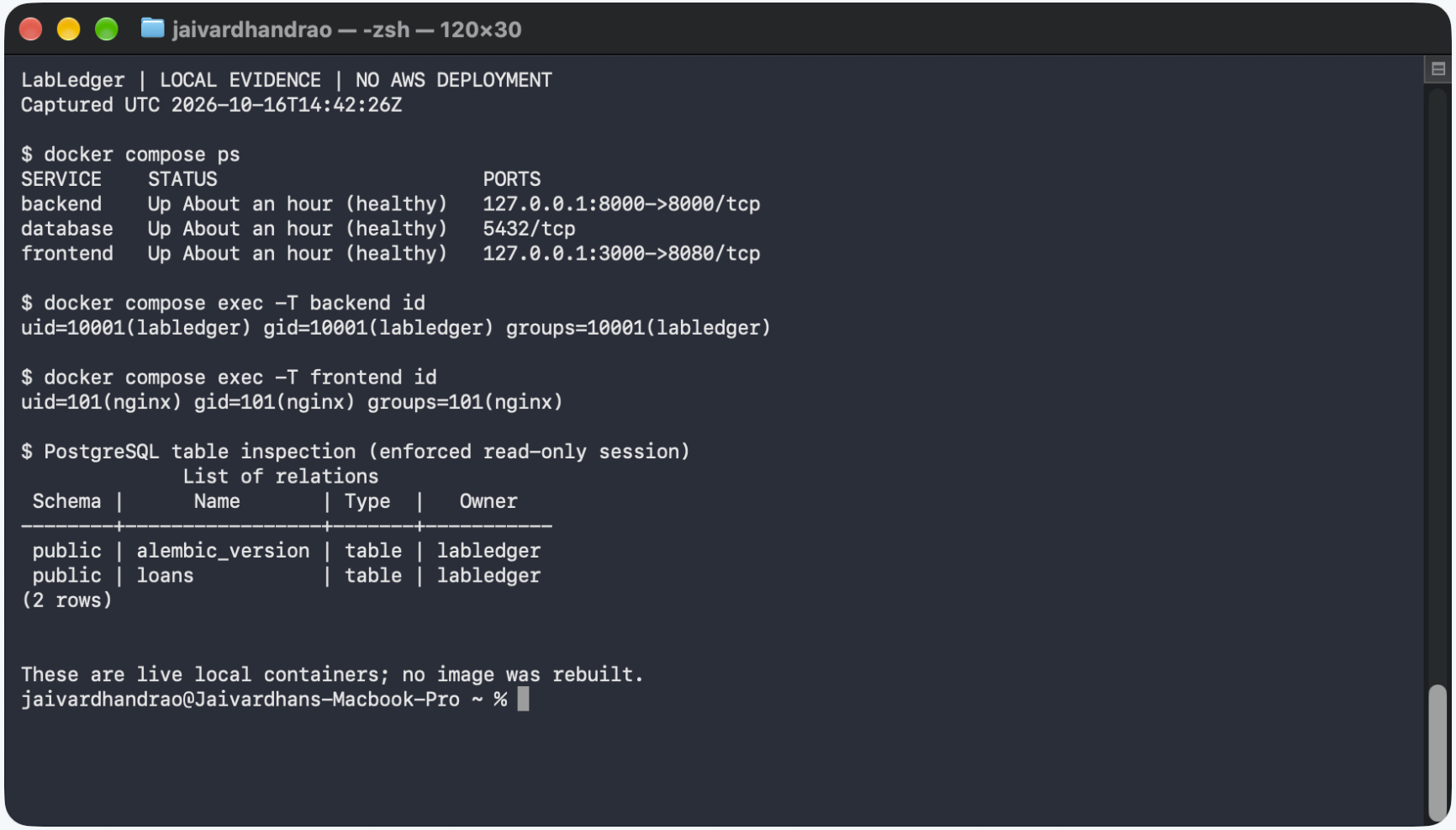
/metrics on both API replicas; real HTTP counters and latency histograms.

35 isolated tests, plus real-stack checks



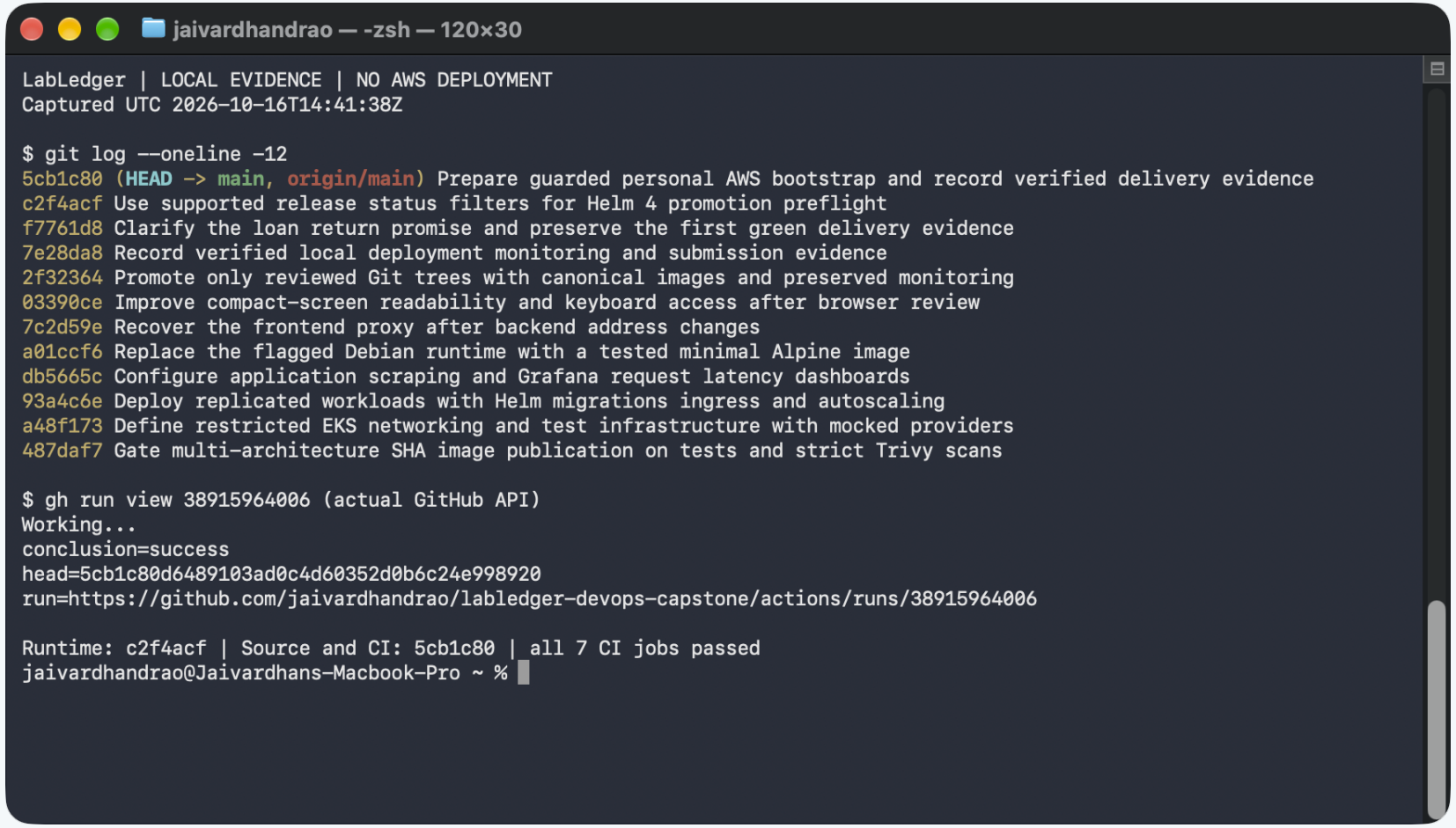
Actual completed CI pytest output; all 35 results are available across the two native captures.

Three healthy services; two non-root images



Live local containers and enforced read-only table inspection; original volumes preserved.

A real change reached the running application



Visible change → green CI → exact c2f4acf local rollout. Current source/CI: 5cb1c80.

Open the actual seven-job green run

The scan gate found a problem, then verified the repair

Before

Strict scan failed

Debian runtime findings were preserved in the original report.

After

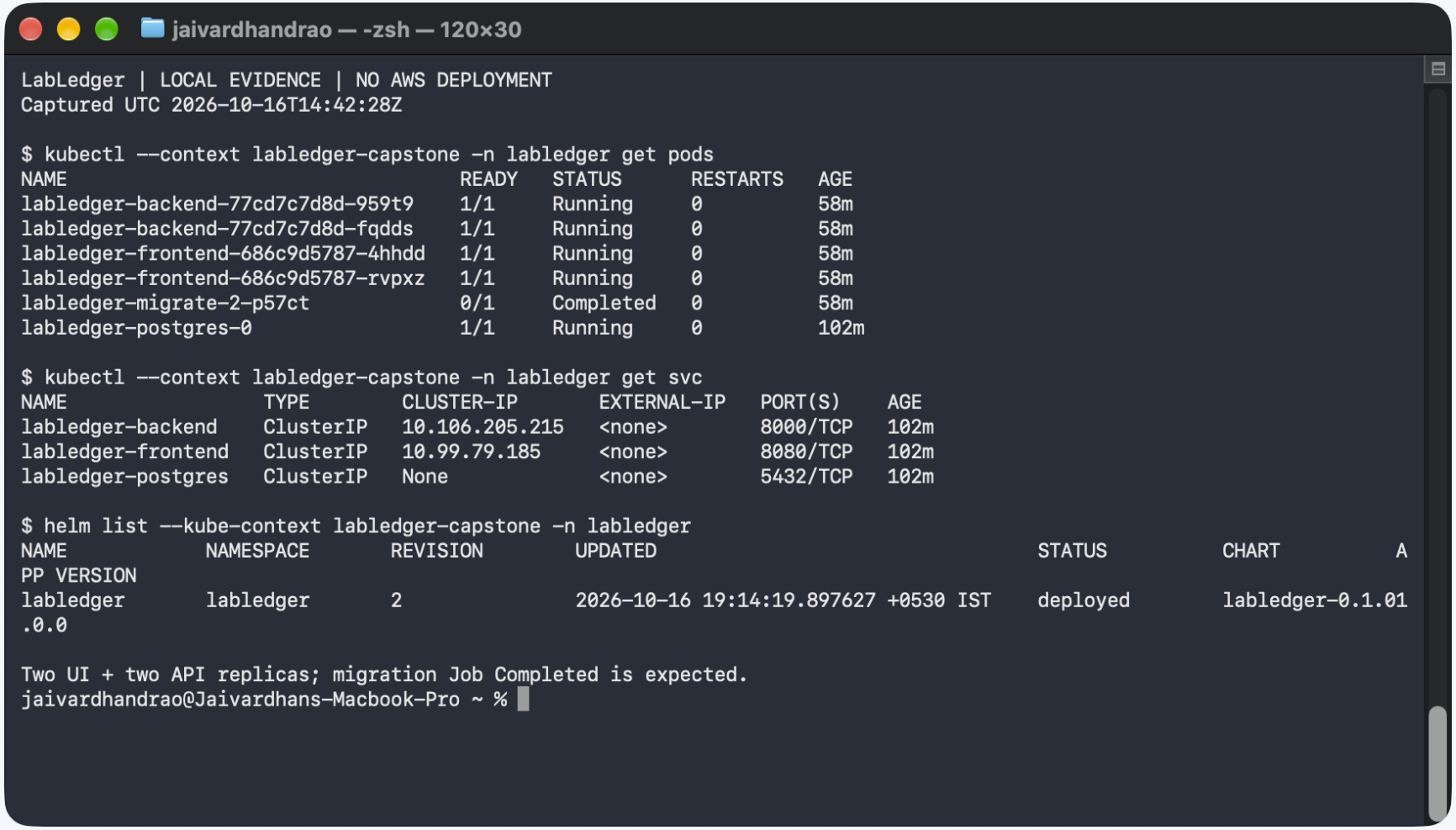
4 clean CI reports

Backend + frontend
amd64 + arm64
0 HIGH / CRITICAL

Gate: --severity HIGH,CRITICAL --ignore-unfixed=false --exit-code 1

A clean scan is evidence about that image at that time; it is not a security guarantee.

Helm release 2 is running in the dedicated local cluster



Local context: labledger-capstone. UI/API Running; migration Job Completed as designed.

Resilience was exercised, not only configured

2 → 4 Ready → 2

Observed API HPA response under controlled local CPU load.

2,338 reads · 0 failures

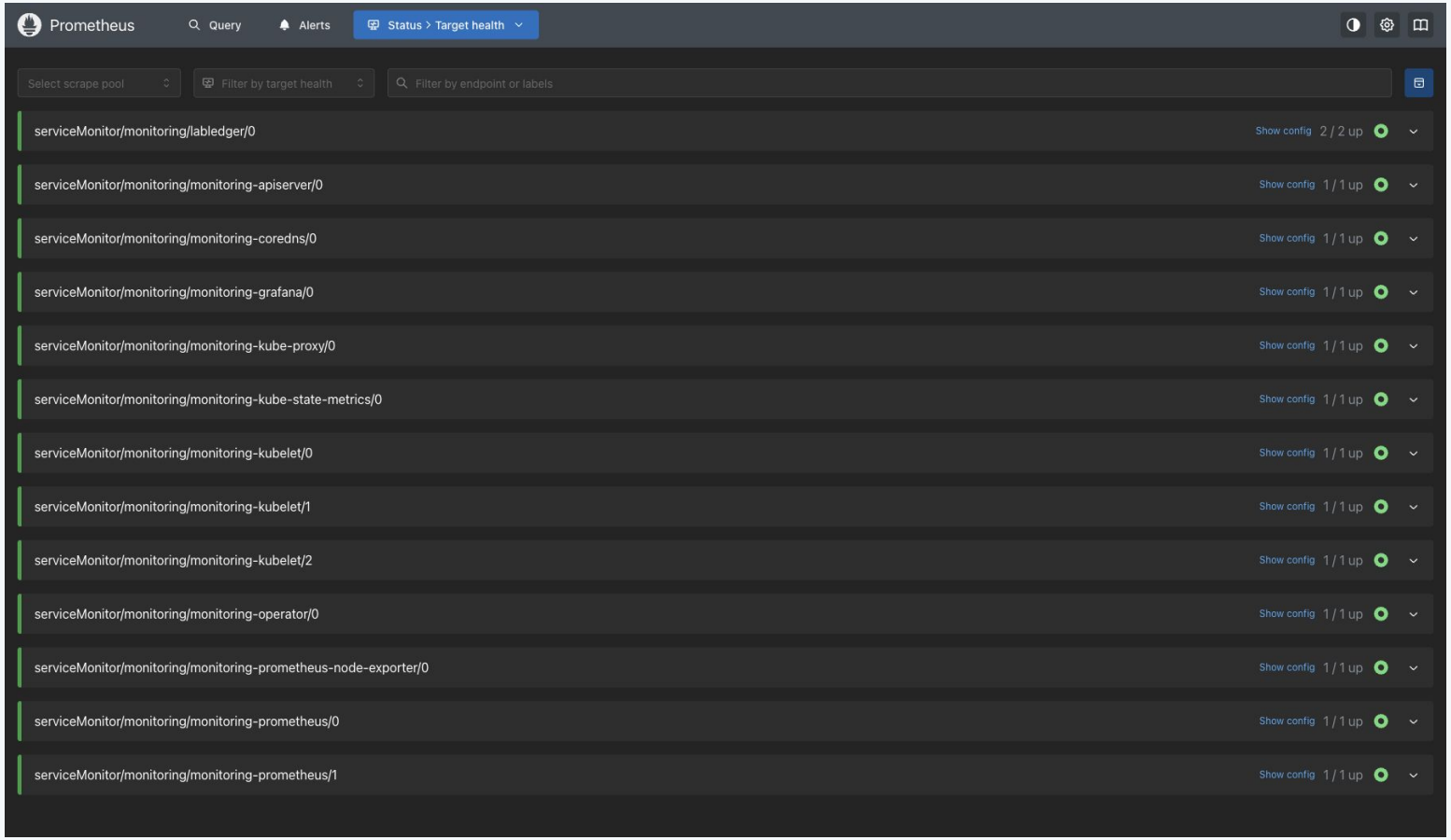
Persistent HTTP connections; original request log retained.

200 → 503 → 200

Service selector fault, diagnosis and real recovery.

Original logs: evidence/local-cluster/README.md. No new failure was injected for these slides.

Actual application scraping is healthy



14 applicable targets UP; original loopback-endpoint failures and local fix retained.

Grafana: real API data verified; populated browser-panel capture still pending login.

AWS is prepared, but has not been deployed

Observed: a real read-only account plan

62 creates · 0 updates · 0 deletes. No apply, access grant or cloud teardown.

Prepared: reproducible infrastructure and bootstrap

VPC + EKS + one worker + NLB; restricted API; Helm and monitoring; project-only cleanup.

Remaining: approval and genuine cloud evidence

Exact access/visibility decisions, actual apply, Console/runtime verification and complete teardown.

Pricing scenario: about US$2.43 before tax. Conditional ceiling: US$10 / maximum 2 hours.

Present the verified work, then review the submission

Ready to review

Original app, tests, containers, green CI, local Kubernetes, native evidence and these editable slides.

Still required for the complete presentation

Live browser/Swagger CRUD, populated Grafana and package-page captures; genuine AWS evidence after approval.

Final submission stays with the student

Form filled for Group A / roll 10117. Review the public repository and evidence before pressing Submit.

Public repository and complete evidence index